



Problem Challenge 2

We'll cover the following



- Rearrange a LinkedList (medium)
- Try it yourself

Rearrange a LinkedList (medium)

Given the head of a Singly LinkedList, write a method to modify the LinkedList such that the **nodes from the second half of the LinkedList are inserted alternately to the nodes from the first half in reverse order**. So if the LinkedList has nodes 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> null, your method should return 1 -> 6 -> 2 -> 5 -> 3 -> 4 -> null.

Your algorithm should not use any extra space and the input LinkedList should be modified in-place.

Example 1:

```
Input: 2 -> 4 -> 6 -> 8 -> 10 -> 12 -> null
Output: 2 -> 12 -> 4 -> 10 -> 6 -> 8 -> null
```

Example 2:

```
Input: 2 -> 4 -> 6 -> 8 -> 10 -> null
Output: 2 -> 10 -> 4 -> 8 -> 6 -> null
```

Try it yourself

Try solving this question here:





```
1
2 class ListNode {
3     int value = 0;
4     ListNode next;
5
6     ListNode(int value) {
7         this.value = value;
8     }
9 }
10
11 class RearrangeList {
12
13     public static void reorder(ListNode head) {
14         // TODO: Write your code here
15     }
16
17     public static void main(String[] args) {
18         ListNode head = new ListNode(2);
19         head.next = new ListNode(4);
20         head.next.next = new ListNode(6);
21         head.next.next.next = new ListNode(8);
22         head.next.next.next.next = new ListNode(10);
23         head.next.next.next.next.next = new ListNode(12);
24         RearrangeList.reorder(head);
25         while (head != null) {
26             System.out.print(head.value + " ");
27             head = head.next;
28         }
29     }
30 }
```

[Run](#)[Save](#)[Reset](#)[← Back](#)[✓ Completed](#)[Next →](#)[Solution Review: Problem Challenge 1](#)[Solution Review: Problem Challenge 2](#)